

MOHAMMED ZAIN RAFEEQUE

Email: zainrafeeqe@gmail.com | Contact: +971-56 957 0799

LinkedIn: linkedin.com/in/zain-rafeeqe/ | GitHub: github.com/ZainRafeeqe

Portfolio: portfoliozain-cwg6.vercel.app



Summary

AI Engineer specialized in production RAG over large multilingual corpora - most recently shipped a bilingual (English + Farsi) system over 247,000+ document chunks at MTA Investment Dubai (~85% retrieval hit rate, 1-5s latency). I focus on systems that ship: measured retrieval quality, deterministic test mode, and observability built in from day one. Stack: Python, LangChain, vLLM, FastAPI, pgvector, ChromaDB, Docker, GitHub Actions. LLM platforms: Claude, OpenAI, Groq, Gemini Vision. DCAI-certified, UAE Resident, available immediately.

Education

Bachelor of Technology in Artificial Intelligence and Data Science | Vimal Jyothi Engineering College

2020 - 2024

Approved by AICTE, affiliated with APJ Abdul Kalam Technological University.

Experience

AI Engineer | MTA Investment LLC, Dubai | Oct 2025 - Apr 2026

- Architected a production-grade bilingual (English + Farsi) RAG expert system for glass manufacturing engineers, combining a fine-tuned Qwen2.5-14B LLM with a hybrid retrieval pipeline (pgvector + BM25 + RRF, cross-encoder reranking) over 247K+ document chunks.
- Delivered the full stack with a React 18 + TypeScript chat UI and Redis semantic caching, achieving ~85% retrieval hit rate and 1-5s end-to-end latency.
- Engineered deployed autonomous agentic RAG retrieval pipelines using ChromaDB, MySQL, and Google Drive for intelligent data querying across internal business systems.
- Built end-to-end automation flows in n8n to streamline complex business processes, deployed Gemini Vision API via Manus AI to extract data from multi-lingual scanned PDFs, and developed custom APIs that connected Manus AI to internal databases - simplifying AI access for non-technical users.

AI Engineer (Internship) | Direct Axis Technologies, Dubai | May 2025 - Aug 2025

- Built a Python cheque-processing pipeline combining OpenCV template matching with deep-learning-based OCR to extract payee, amount, date, and signature fields from cheque images.
- Built a Python time-series forecasting pipeline using Facebook Prophet - JSON-based data I/O, configurable seasonality and changepoint controls, MAE/MAPE/RMSE evaluation, cross-validation, and grid-search hyperparameter tuning for financial-workflow forecasts.

AI/ML Engineer (Training Program) | Meta Scifor Technologies | Oct 2024 - Mar 2025

GitHub: github.com/ZainRafeeqe/AI-Learning-Style-Predictor

- Completed a remote AI/ML training program covering machine-learning algorithms, deep-learning frameworks, and Python engineering.
- Capstone project: built an ML pipeline classifying students into VARK learning styles - lifted accuracy from 59.2% baseline to 84.5% (+25.3 pp) with engineered features and a Streamlit demo.

Projects

Wakeel - Domain-Specific Legal AI Platform (UAE) | May 2026

GitHub: github.com/ZainRafeeqe/Wakeel

- Designed and built a domain-specific legal AI platform - extends classic RAG with semantic lawyer matching (MiniLM embeddings + cosine similarity), intake parsing via Gemini Vision OCR + Llama-3.3-70B structured extraction, and a consultation booking flow, beyond document Q&A.
- Shipped a hermetic RAG evaluation harness on 30 hand-curated UAE-law questions - Hit@5 = 100%, Mean Reciprocal Rank = 1.000, keyword faithfulness = 98.3%, median retrieval latency 7.2 ms (p95: 9.7 ms) - runs deterministically without API keys.
- Built reliability scaffolding throughout: per-request UUID tracing via ContextVar, JSON structured logging with token-usage + latency per LLM call, retry-with-exponential-backoff, graceful mock-mode fallback, full pytest suite.

DocIntel - Full-Stack PDF Intelligence Platform | May 2026

GitHub: github.com/ZainRafeeqe/DocIntel | **Live:** doc-intel-delta.vercel.app

- Designed and deployed a full-stack RAG platform enabling chat over PDFs with page-level citations and structured JSON extraction, handling PDFs from 5 to 500+ pages without performance degradation.
- Combined dense (sentence-transformers MiniLM-L6-v2) + lexical (BM25Okapi) retrieval via Reciprocal Rank Fusion (k=60), served by Groq Llama-3.3-70B-versatile for both Q&A and JSON-mode structured extraction with one-click CSV download.
- Deployed full-stack on Render + Vercel with the embedding model baked into the Docker image to survive cold starts.

Agentic RAG System - Multi-Agent Architecture | Jan 2026

GitHub: github.com/ZainRafeeqe/Agentic_rag

- Built a 4-agent retrieval pipeline (Router -> Planner -> Executor -> Synthesizer) over a ChromaDB vector store, served via FastAPI with sentence-transformers embeddings and an Ollama-backed local LLM.
- Extended the system with a ConnectorFactory pattern and 5 source connectors - local files (PDF/DOCX/TXT/CSV), PostgreSQL, MySQL, REST APIs, Google Drive - all yielding a unified Document interface.
- Built an IngestionAgent orchestrating extract -> sentence-aware chunk -> embed -> store into ChromaDB; configurable chunk size / overlap, batched writes, IngestionReport summaries, 15 tests.

Publications

A Novel Approach to Malayalam Speech-to-Text and Text-to-English Translation | April 2024

Research project on offline neural machine translation between Malayalam and English using IndicTrans2 + ctranslate2. Project report and certificate of publication available at github.com/ZainRafeeqe/Malayalam-speech-translation.

Skills

- **Core:** Python, LangChain, RAG, Fine-tuning, vLLM, FastAPI, pgvector, MCP
- **AI / ML:** LLMs, LangGraph, Prompt Engineering, sentence-transformers, Hugging Face, OpenCV, Prophet, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch
- **LLM Platforms:** Claude (Anthropic), Claude Code, OpenAI, Groq, Gemini Vision, Manus AI
- **Databases:** PostgreSQL, pgvector, MySQL, ChromaDB, Redis
- **Backend & DevOps:** Docker, GitHub Actions (CI/CD), FastAPI, Nginx, Git, n8n
- **Frontend:** Next.js 14, React 18, TypeScript, Tailwind, Streamlit, Gradio
- **Languages:** Python, SQL, Java, R

Certifications

- **One Million Prompters (AI Prompt Engineering)** - Dubai Centre for Artificial Intelligence (DCAI) | 2025